

Problem

Submissions

Leaderboard

Discussions

Given two sorted arrays, merge them and find the median of the combined sorted array.

Input Format

First line contains two integers n and m. Second line contains n sorted integers. Third line contains m sorted integers.

Constraints

$1 \leq n, m \leq 10^5$ $-10^6 \leq \text{array elements} \leq 10^6$

Output Format

Print the median of the merged array.

If the total number of elements is odd, print the middle element. If even, print the average of the two middle elements with one decimal place.

Sample Input 0

```
2 2
1 2
3 4
```

Sample Output 0

```
2.5
```

Sample Input 1

```
2 3
1 2
3 4 5
```

Sample Output 1

```
3.0
```

Sample Input 2

```
4 2
1 3 5 7
2 4
```

Sample Output 2

```
3.5
```

Change Theme

Language Python 3

↺

⋮

```
1 # Enter your code here. Read input from STDIN.
2
3
4 n, m = list(map(int, input().split()))
5
6 arr1 = list(map(int, input().split()))
7 arr2 = list(map(int, input().split()))
8
9 arr = arr1 + arr2
10 arr.sort()
11
12 length= n + m
13 if length%2 !=0:
14     i = length//2
15     avg = arr[i]
16 else:
17     avg = (arr[length//2 - 1] + arr[length//2])
18 print(f"{avg:.1f}")
```

Line: 18 Col: 20

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☐ Test against custom input

✔ Test case 0

✔ Test case 1

✔ Test case 2

Compiler Message

Success

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